

K.S.R.M. COLLEGE OF ENGINEERING (AUTONOMOUS), KADAPA
B.Tech IV Semester (R23UG) Reg. & Supple. Exams Jun. – Jul. 2025
OPERATING SYSTEMS
(CSE)

Time: 3 Hours**Max. Marks: 70****Instructions:**

1. Answer all 10 questions from Part-A. Each question carries two marks
2. Answer one full question from each unit in Part-B. Each full question carries 10 marks

PART-A				10*2 = 20M		
1	a	What are system calls? List different types of system calls.	2M	CO1	L2	
	b	What are the key services provided by an operating system?	2M	CO1	L2	
	c	What is the difference between a process and a thread?	2M	CO2	L2	
	d	Explain the purpose of a Process Control Block (PCB).	2M	CO2	L2	
	e	Define deadlock with example.	2M	CO3	L2	
	f	Define mutex locks	2M	CO3	L2	
	g	List different page replacement algorithms	2M	CO4	L2	
	h	What is the use of Translation Lookaside Buffer (TLB)?	2M	CO4	L2	
	i	What is an access matrix?	2M	CO5	L2	
	j	List the different file operations	2M	CO5	L2	
PART-B				5*10=50M		
UNIT-I						
2		Define operating system. Explain the various functions of operating systems.	10M	CO1	L2	
OR						
3	a	List and explain different types of system programs with examples.	5M	CO1	L2	
	b	Illustrate different Operating Systems Operations	5M	CO1	L3	
UNIT-II						
4	a	Explain the different states of a process with a diagram.	5M	CO2	L2	
	b	Describe the First Come First Serve (FCFS) scheduling algorithm with an example.	5M	CO2	L3	
(OR)						
5	a	Define multithreading and discuss different multithreading models.	5M	CO2	L1	
	b	Discuss multiple processor scheduling	5M	CO2	L2	
UNIT-III						
6	a	Define the critical section problem and explain its significance.	5M	CO3	L1	
	b	What is a semaphore? Explain its use in process synchronization.	5M	CO3	L2	

(OR)					
7	a	Explain the four necessary conditions for deadlock	5M	CO3	L2
	b	Describe different methods for handling deadlocks.	5M	CO3	L2
UNIT-IV					
8	a	Explain about paging concept.	5M	CO4	L2
	b	Explain about demand paging	5M	CO4	L2
(OR)					
9		Explain FIFO, OPR and LRU page replacement algorithms. Consider page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 and 3 frames in main memory. Find the number of page faults for the page replacement algorithms FIFO, OPR and LRU.	10M	CO4	L3
UNIT-V					
10	a	Explain different file accessing methods.	5M	CO5	L2
	b	What is protection? Explain goals, principles and domain of protection	5M	CO5	L2
(OR)					
11	a	What is a directory? Explain different directory structures.	5M	CO5	L2
	b	Explain different types of security threats in an operating system	5M	CO5	L2
